Brain Attack Pathway (BAT): June 2023

(Not to be a substitution for proper clinical judgment)

Any child with acute focal neurological deficit or altered mental status

Screening Questions (Needs 1):
- Unilateral Weakness
- Unilateral Numbness
- Facial Droop (sparing forehead)
- Aphasia or Dysarthria
- Unilateral incoordination
- Visual loss or Double Vision

Alert Criteria (Needs ALL 3):
1. Symptom Onset < 12 hours
2. No prior hx of Todd's paralysis
3. Problem began or worsened suddenly

Further History:
- Time of last known normal
- NPO status
- Hx of headache, seizure, URI, sickle cell, cardiac disease, neck trauma, clotting disorder
- Family hx of young stroke
- Home meds (including OCPs)

If 1 screening question AND all 3 alert criteria met → NIHSS + Page Neurology to discuss activating BAT

Step 1a

Initial management
1. ABCs, CR monitor, make NPO, place 2 large bore IVs
2. Glucose stick, CBC, PT/PTT/INR, type & screen, CRP/ESR, urine drug screen, pregnancy test (female > 9 y/o), EKG
3. HOB flat, strict bed rest

Neuroprotective measures
1. Normovolemia: Isotonic fluid (0.9% NS) at maintenance w/ prn bolus
2. Normotension: SBP between 50-90th % for age, treat low BPs w/ NS ± pressors; tx htn with labetalol to lower by 25% over 24 hours
3. Normothermia: Treat T >38 C with rectal acetaminophen
4. Normoglycemia: No glucose in IVF unless hypoglycemic
5. Normal O2 and CO2: > 92% O2
6. Seizure control: AEDs ASAP with any seizure activity

Step 1b

NIHSS by certified provider (*ED & PICU fellows & attendings*)
STAT CT Head without contrast

→ Bleed/Tumor? →
1. Reverse Coagulopathy
2. Call Neurosurgery

Step 2

Neuro – Reassess need to continue pathway with primary
ED/PICU – Call radiology tech, pharmacy (x7273: ASK FOR ICU RPH) and anesthesia if needed (place note in chart to document emergent status)

If at OUTSIDE HOSPITAL:
1. NORTON Facility: Stat STROKE PROTOCOL head CT
2. If outside Norton stat non-contrast CT of head; send copy of imaging with patient
3. Arrange for transport to NCH:
   Just For Kids transport: 888-729-9111
4. Notify NCH ED of ETA (629-4019)

Step 3

A. Fast brain MRI "Stroke Protocol"
DWI, SWI, Axial FLAIR, Axial T2, Sag T1; MRA head & neck; MR perfusion

B. If sedated, consider additional imaging options that may help

- **"Fat Sat"** if hx of neck trauma/pain
- **MRV** if presenting w/headache, chemotherapy, pregnancy, papilledema
- **Contrast**, if concern for tumor, demyelination, or infection
- **Case dependent** – consider CTA/CTP "CT stroke protocol"

Step 4

No ischemic stroke
- PICU vs floor
- Neurologic workup

Ischemic stroke w/open vessels
- Acute stroke management in PICU
- Neuroprotective measures
- Aspirin – If >15kg, ASA 81 mg once. If <15kg, ASA 5 mg/kg
- Etiology: Hypercoagulable w/up w/hematology, TTE w/bubble study

Ischemic stroke w/concern for dissection or multiple emboli
- Acute stroke management in PICU
- Neuroprotective measures
- ±Heparin w/heme/cardio consult
- Etiology: Hypercoagulable w/up, TTE w/bubble study vs TEE

Ischemic stroke w/large vessel occlusion (M1, ICA, Basilar)
- Neuroprotective measures
- Review Alteplase checklist
- Discuss w/neurovascular
- Etiology work up

Step 5

IV Alteplase Criteria: Onset < 4.5 hours, NIHSS > 6 and < 24, no exclusions to IV Alteplase, MRI appropriate, signed Alteplase consent; If yes, go to page 2

Clot Retrieval Criteria: Onset < 12 hours for anterior, or <24 hours for posterior, must be deemed appropriate by neuro intervention

Acute Alteplase Management for Stroke Pathway: June 2023

Contraindications to Alteplase

History
1. > 4.5 hours from last know normal
2. Time of symptoms start unknown
3. Stroke, major head trauma, or intracranial surgery in last 3 months
4. History of ICH, aneurysm, or AVM
5. Major surgery in last 10 days
6. GI or GU bleeding in last 21 days
7. Hx of neoplasm or malignancy
8. Underlying significant bleeding disorder
9. Dx of PACNS or secondary angitis
10. Seizure at onset (relative contraindication)
11. Heparin administration within 4 hours and elevated aPTT
12. DOAC administration within 48 hours
13. Treatment dose enoxaparin administration within 24 hours

Patient factors
Would decline blood transfusion if needed
Presentation consistent with acute MI or pericarditis
Arterial puncture at non compressible site or LP within past 7 days

Etiology:
Stroke due to SBE, sickle cell disease, moyamoya disease, meningitis, embolism (other than thrombotic), MELAS

Exam:
Persistent SBP > 15% above 95th % for age
Peds NIHSS < 6 at the time beginning infusion
Resolving deficit without intervention

Imaging:
ICH on CT or MRI or symptoms suggesting SAH (even CT negative)
Hypodensity on CTH
> 1/3 MCA territory involvement on CT or MRI
Intracranial cerviocephalic dissection

Labs:
D-stick < 50 mg/dl or > 400 mg/dl
Bleeding diathesis, platelets<100,000, PT > 15 or INR > 1.4

Confirmed IV Alteplase candidate
Pharmacy prepares Alteplase infusion and delivers to ED or ICU
Total dose – 0.9 mg/kg IV
Bolus dose – 10% of total dose IV over 5 mins
Infusion dose - remaining 90% IV over 1 hour
Nurse/MD double checks dose with pharmacy

Treat to lower BP if > 15% above 95% for age for > 1 hour or > 20% at any time

Maintain cardiorespiratory and BP monitoring
If age based parameters acceptable, give Alteplase when available
Systolic BPs should be maintained between 50th % for age and 15% above 95% for age

Give labetalol 0.2 mg/kg IV push over 2-3 min, repeat q15 mins prn; consider nicardipine drip, 1 mcg/kg/min, titrate to desired BP

During Alteplase

Q15 minute vital signs
Monitor for acute neuro changes (HA, seizure, N/V, acute HTN)

If neuro changes
1. Stat non-contrast head CT
2. If positive for hemorrhage:
Prepare to give cryoprecipitate (containing factor VIII) (1-2 U/10 kgs body weight; 5 mls/kg is less than 30 kgs)
Consider Tranexamic acid (TXA) 15mg/kg IV if cryoprecipitate is delayed
(See alteplase reversal included in anticoagulant reversal order set)

Post Alteplase

Vitals + Neuro-checks Q15 mins for 2 hours Q30 mins for 6 hours Q1h for 24 hours

No foley for 4 hours
No NGT for 24 hours
No anticoagulants or antithrombotics for 24 hours
Repeat CTH for 24 hours

Systolic Blood Pressure Parameters - Male

Age	50%	95%	> 15% above 95%	> 20% above 95%
1-4 years	90	112	129	134
5 years	95	113	130	136
6-10 years	96	121	139	145
11-18 years	105	140	161	168
>18 years	110	140	161	168

Systolic Blood Pressure Parameters - Female

Age	50%	95%	>15% above 95%	> 20% above 95%
1-4 years	90	111	128	133
5 years	94	113	130	136
6-10 years	96	121	139	145
11-18 years	105	131	151	157
>18 years	110	140	161	168